Manideep_stat_proj

2025-05-10

```r
library(data.table)
library(ggplot2)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:data.table':
##
##     between, first, last

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
dt <- fread("data.csv")

dt[, race   := factor(race)]
dt[, gender := factor(gender)]
dt[, fgen   := factor(fgen, levels=c(0,1), labels=c("NonFirstGen","FirstGen"))]
dt[, urban  := factor(urban)]

dt[, race_num  := as.integer(race)]
dt[, gender01  := as.integer(gender == levels(gender)[2])]
dt[, fgen01    := as.integer(fgen == "FirstGen")]
dt[, urban_num := as.integer(urban)]

str(dt)
```

```
## Classes 'data.table' and 'data.frame':   10391 obs. of  17 variables:
##  $ y        : num  -0.3655 -0.1591 -0.2554 0.1909 -0.0176 ...
##  $ z        : int  1 1 1 1 1 1 1 1 1 1 ...
##  $ selfrpt  : int  6 4 6 6 6 6 6 6 6 6 ...
##  $ race     : Factor w/ 15 levels "1","2","3","4",..: 4 12 4 4 4 4 4 4 5 4 ...
##  $ gender   : Factor w/ 2 levels "1","2": 2 2 2 2 1 2 1 1 2 1 ...
##  $ fgen     : Factor w/ 2 levels "NonFirstGen",..: 2 2 1 1 1 1 1 1 1 2 ...
##  $ urban    : Factor w/ 5 levels "0","1","2","3",..: 5 5 5 5 5 5 5 5 5 5 ...
##  $ mindset  : num  0.335 0.335 0.335 0.335 0.335 ...
```

```
##  $ test     : num  0.649 0.649 0.649 0.649 0.649 ...
##  $ sch_race : num  -1.31 -1.31 -1.31 -1.31 -1.31 ...
##  $ pov      : num  0.224 0.224 0.224 0.224 0.224 ...
##  $ size     : num  -0.427 -0.427 -0.427 -0.427 -0.427 ...
##  $ schoolid : int  76 76 76 76 76 76 76 76 76 76 ...
##  $ race_num : int  4 12 4 4 4 4 4 4 5 4 ...
##  $ gender01 : int  1 1 1 1 0 1 0 0 1 0 ...
##  $ fgen01   : int  1 1 0 0 0 0 0 0 0 1 ...
##  $ urban_num: int  5 5 5 5 5 5 5 5 5 5 ...
##  - attr(*, ".internal.selfref")=<externalptr>
```

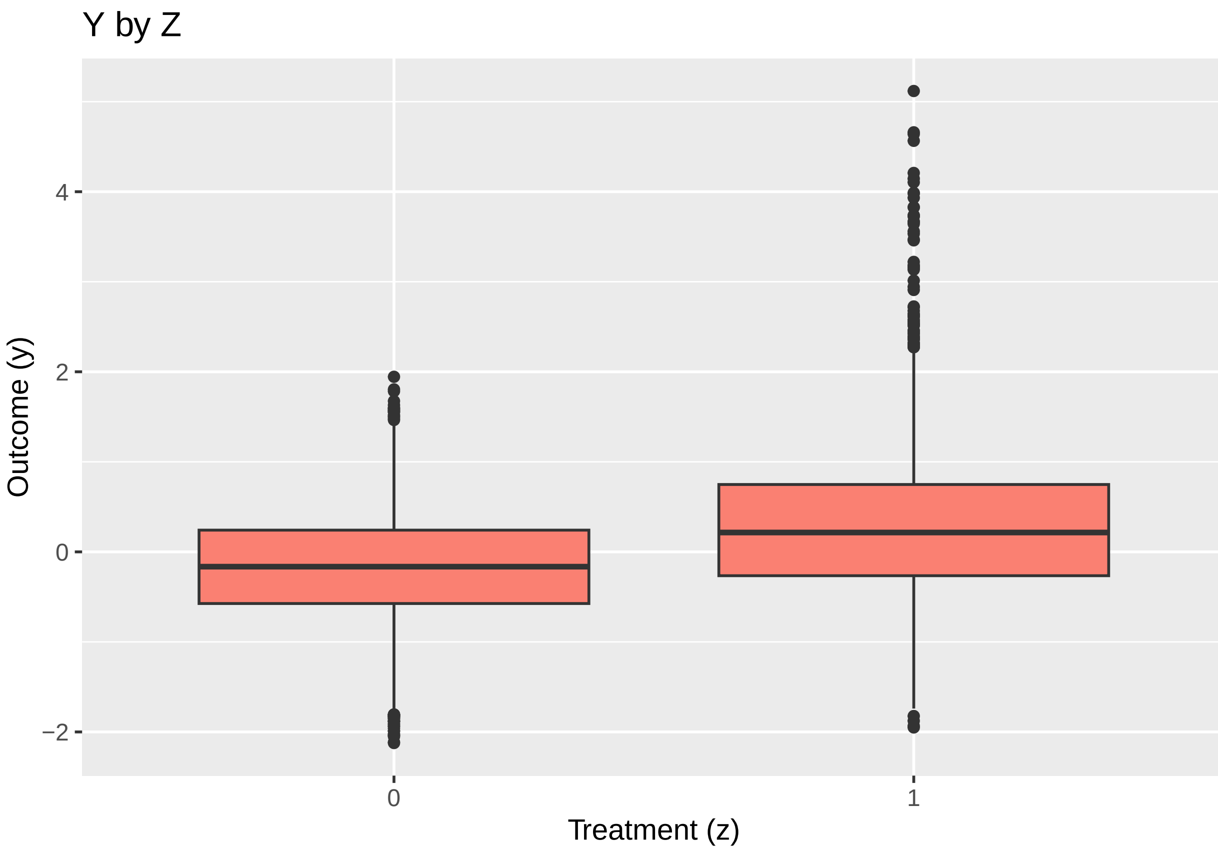
```
cat("Number of rows:   ", nrow(dt), "\n",
    "Number of columns:", ncol(dt), "\n")
```

```
## Number of rows:    10391
##  Number of columns: 17
```

```
num_vars <- c("y","selfrpt","mindset","test","sch_race","pov","size")
dt[, lapply(.SD, function(x) list(mean=mean(x), sd=sd(x), median=median(x))), .SDcols=num_vars]
```

```
##               y  selfrpt      mindset        test    sch_race         pov
##          <list>   <list>       <list>      <list>      <list>      <list>
## 1: -0.02024954 5.268117  -0.04045744  0.05484058  -0.0893489 -0.04591054
## 2:   0.7188819 1.120765    0.9697429   0.9355595   0.9628039   0.9672625
## 3: -0.05644734        5 -0.009953883 -0.02251353 -0.05703584  -0.1596016
##            size
##          <list>
## 1: -0.02616795
## 2:    1.010387
## 3:  -0.2115532
```

```
ggplot(dt, aes(x=factor(z), y=y)) +
  geom_boxplot(fill="salmon") +
  labs(x="Treatment (z)", y="Outcome (y)", title="Y by Z")
```




```
cor_mat <- cor(dt[, ..num_vars])
print(round(cor_mat,2))
```

```
##                y selfrpt mindset  test sch_race   pov  size
## y           1.00    0.38   -0.04 -0.10     0.16  0.04  0.01
## selfrpt     0.38    1.00   -0.05  0.07    -0.08 -0.05  0.05
## mindset    -0.04   -0.05    1.00 -0.57     0.25  0.26 -0.49
## test       -0.10    0.07   -0.57  1.00    -0.48 -0.42  0.48
## sch_race    0.16   -0.08    0.25 -0.48     1.00  0.42 -0.01
## pov         0.04   -0.05    0.26 -0.42     0.42  1.00 -0.12
## size        0.01    0.05   -0.49  0.48    -0.01 -0.12  1.00
```

```
# Simple Additive Linear Regression (Unadjusted)
model_slr <- lm(y ~ z, data = dt)
summary(model_slr)
```

```
##
## Call:
## lm(formula = y ~ z, data = dt)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -2.2381 -0.4535 -0.0144  0.4254  4.8308
##
## Coefficients:
```

```
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept) -0.169049   0.008199  -20.62   <2e-16 ***
## z            0.456907   0.014367   31.80   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6863 on 10389 degrees of freedom
## Multiple R-squared:  0.08872,    Adjusted R-squared:  0.08863
## F-statistic:  1011 on 1 and 10389 DF,  p-value: < 2.2e-16
```

```r
unadj_est <- coef(model_slr)["z"]
unadj_se <- summary(model_slr)$coefficients["z", "Std. Error"]
unadj_ci_lower <- unadj_est - 1.96 * unadj_se
unadj_ci_upper <- unadj_est + 1.96 * unadj_se

cat("Unadjusted ATE Estimate:", round(unadj_est, 4), "\n")
```

```
## Unadjusted ATE Estimate: 0.4569
```

```r
cat("Standard Error:", round(unadj_se, 4), "\n")
```

```
## Standard Error: 0.0144
```

```r
cat("95% CI: [", round(unadj_ci_lower, 4), ", ", round(unadj_ci_upper, 4), "]\n")
```

```
## 95% CI: [ 0.4287 ,  0.4851 ]
```

```r
# Blocked Design
dt[, test_q := factor(ntile(test, 3), labels = paste0("Q",1:3))]
dt[, pov_q := factor(ntile(pov, 3), labels = paste0("Q",1:3))]
dt[, block := interaction(test_q, pov_q, drop = TRUE)]

dt[, p_treated := mean(z), by = block]

dt[, wt_block := ifelse(z == 1, 1/p_treated, 1/(1-p_treated))]

blocked_model <- lm(y ~ z, data = dt, weights = wt_block)
blocked_effect <- coef(blocked_model)["z"]
blocked_se <- summary(blocked_model)$coefficients["z", "Std. Error"]
blocked_ci_lower <- blocked_effect - 1.96 * blocked_se
blocked_ci_upper <- blocked_effect + 1.96 * blocked_se

cat("Blocked Design ATE Estimate:", round(blocked_effect, 4), "\n")
```

```
## Blocked Design ATE Estimate: 0.4729
```

```r
cat("Standard Error:", round(blocked_se, 4), "\n")
```

```
## Standard Error: 0.0143
```

```r
cat("95% CI: [", round(blocked_ci_lower, 4), ", ", round(blocked_ci_upper, 4), "]\n")
```

```
## 95% CI: [ 0.4448 ,  0.5009 ]
```

```r
# AIPW (Double-Dipping)
library(boot)

prop_model <- glm(z ~ selfrpt + mindset + test + sch_race + pov + size +
                   race + gender + urban + fgen,
                 family = binomial(), data = dt)
dt[, ps := predict(prop_model, type = "response")]
dt[, ps_trim := pmin(pmax(ps, 0.01), 0.99)]

dt[, wt_aipw := ifelse(z == 1, 1/ps_trim, 1/(1-ps_trim))]

om1f <- lm(y ~ selfrpt + race + gender + fgen + urban + mindset + test + sch_race + pov + size,
           data=dt[dt$z==1,])
om0f <- lm(y ~ selfrpt + race + gender + fgen + urban + mindset + test + sch_race + pov + size,
           data=dt[dt$z==0,])

mu1f <- predict(om1f, newdata = dt)
mu0f <- predict(om0f, newdata = dt)

dr1f <- mu1f + (dt$z==1)*(dt$y - mu1f)/dt$ps_trim
dr0f <- mu0f + (dt$z==0)*(dt$y - mu0f)/(1-dt$ps_trim)
aipw_est <- mean(dr1f) - mean(dr0f)

boot_aipw <- boot(dt, function(d,i){
  db  <- d[i, ]
  m   <- glm(z ~ selfrpt + race + gender + fgen + urban + mindset + test + sch_race + pov + size,
            data=db, family=binomial)
  p   <- m$fitted.values
  p   <- pmin(pmax(p, 0.01), 0.99)
  o1 <- lm(y ~ selfrpt + race + gender + fgen + urban + mindset + test + sch_race + pov + size,
           data=db[db$z==1,])
  o0 <- lm(y ~ selfrpt + race + gender + fgen + urban + mindset + test + sch_race + pov + size,
           data=db[db$z==0,])
  m1 <- predict(o1, newdata=db)
  m0 <- predict(o0, newdata=db)
  d1 <- m1 + (db$z==1)*(db$y - m1)/p
  d0 <- m0 + (db$z==0)*(db$y - m0)/(1-p)
  mean(d1) - mean(d0)
}, R=1000)

aipw_se <- sd(boot_aipw$t)
aipw_ci <- quantile(boot_aipw$t, c(0.025, 0.975))

cat("AIPW ATE Estimate:", round(aipw_est, 4), "\n")
```

```
## AIPW ATE Estimate: 0.4118
```

```r
cat("Bootstrap Standard Error:", round(aipw_se, 4), "\n")
```

```
## Bootstrap Standard Error: 0.0137
```

```r
cat("95% Bootstrap CI: [", round(aipw_ci[1], 4), ", ", round(aipw_ci[2], 4), "]\n")
```

```
## 95% Bootstrap CI: [ 0.3856 ,  0.4407 ]
```

```r
# Double Machine Learning (DML)
library(ranger)
library(glmnet)
```

```
## Loading required package: Matrix
```

```
## Loaded glmnet 4.1-8
```

```r
library(SuperLearner)
```

```
## Loading required package: nnls
```

```
## Loading required package: gam
```

```
## Loading required package: splines
```

```
## Loading required package: foreach
```

```
## Loaded gam 1.22-5
```

```
## Super Learner
```

```
## Version: 2.0-29
```

```
## Package created on 2024-02-06
```

```r
library(boot)

K <- 5

set.seed(42)
dt[, fold := sample(1:K, .N, replace = TRUE)]

dt[, y_resid := NA_real_]
dt[, z_resid := NA_real_]

covariates <- c("selfrpt", "mindset", "test", "sch_race", "pov", "size",
                "race_num", "gender01", "urban_num", "fgen01")

outcome_formula <- as.formula(paste("y ~", paste(covariates, collapse = " + ")))
```

```r
treatment_formula <- as.formula(paste("z ~", paste(covariates, collapse = " + ")))

create_matrix <- function(dt_subset, formula) {
  model_matrix <- model.matrix(formula, data = dt_subset)
  return(model_matrix[, -1])
}

for (k in 1:K) {
  train_data <- dt[fold != k]
  test_data <- dt[fold == k]

  X_train <- create_matrix(train_data, ~ selfrpt + mindset + test + sch_race + pov +
                             size + race_num + gender01 + urban_num + fgen01)
  X_test <- create_matrix(test_data, ~ selfrpt + mindset + test + sch_race + pov +
                            size + race_num + gender01 + urban_num + fgen01)

  z_model <- cv.glmnet(X_train, train_data$z, family = "binomial", alpha = 0.5)
  z_pred <- predict(z_model, newx = X_test, type = "response", s = "lambda.min")[,1]
  dt[fold == k, z_resid := z - z_pred]

  y_model <- ranger(outcome_formula, data = train_data,
                    num.trees = 500, seed = 42, min.node.size = 5)
  y_pred <- predict(y_model, data = test_data)$predictions
  dt[fold == k, y_resid := y - y_pred]
}

dml_model <- lm(y_resid ~ z_resid - 1, data = dt)
dml_est <- coef(dml_model)[1]
dml_se <- summary(dml_model)$coefficients[1, 2]
dml_ci_lower <- dml_est - 1.96 * dml_se
dml_ci_upper <- dml_est + 1.96 * dml_se

alpha <- 0.05
t_value <- dml_est / dml_se
p_value <- 2 * pt(-abs(t_value), df = nrow(dt) - 1)
reject_h0 <- p_value < alpha

dt[, wt_dml := z_resid]

cat("DML ATE Estimate:", round(dml_est, 4), "\n")
```

```
## DML ATE Estimate: 0.4175
```

```r
cat("Standard Error:", round(dml_se, 4), "\n")
```

```
## Standard Error: 0.0123
```

```r
cat("95% CI: [", round(dml_ci_lower, 4), ", ", round(dml_ci_upper, 4), "]\n")
```

```
## 95% CI: [ 0.3934 ,  0.4416 ]
```

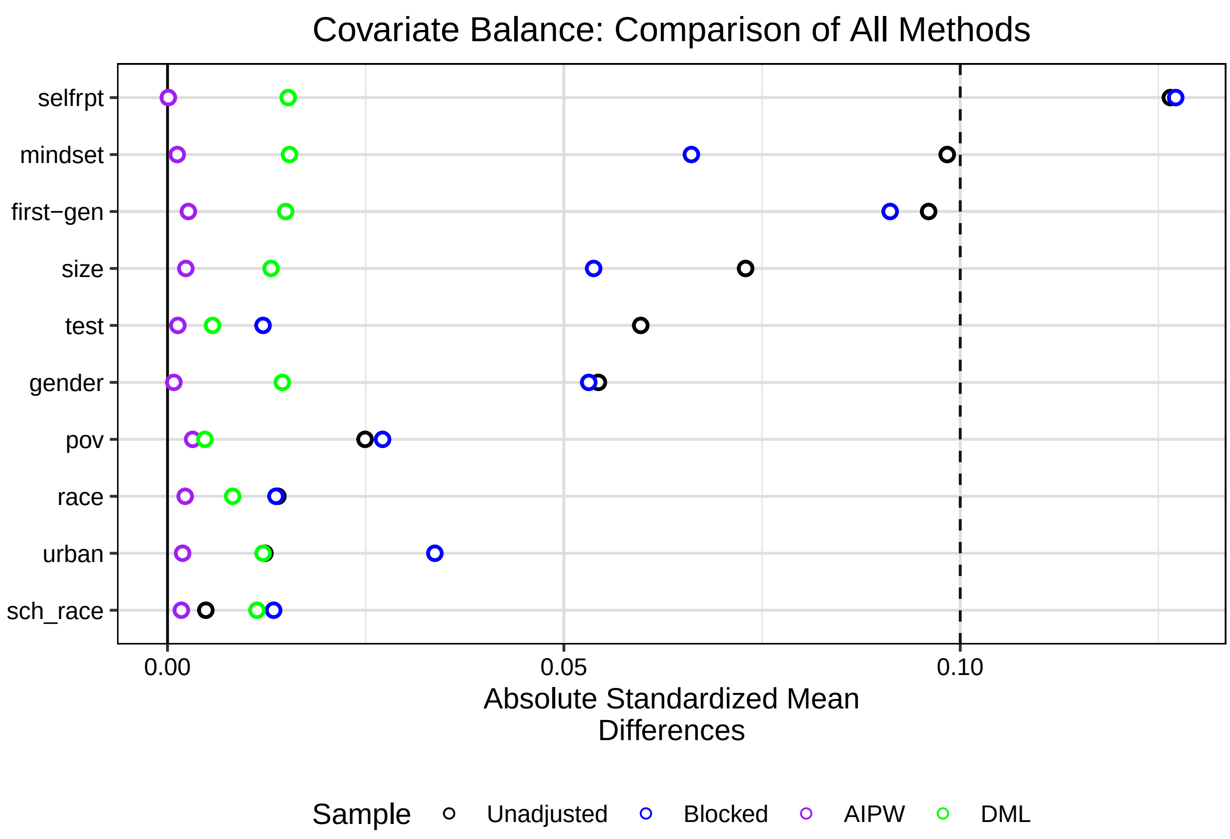
```r
cat("p-value:", round(p_value, 4), "\n")
```

```
## p-value: 0
```

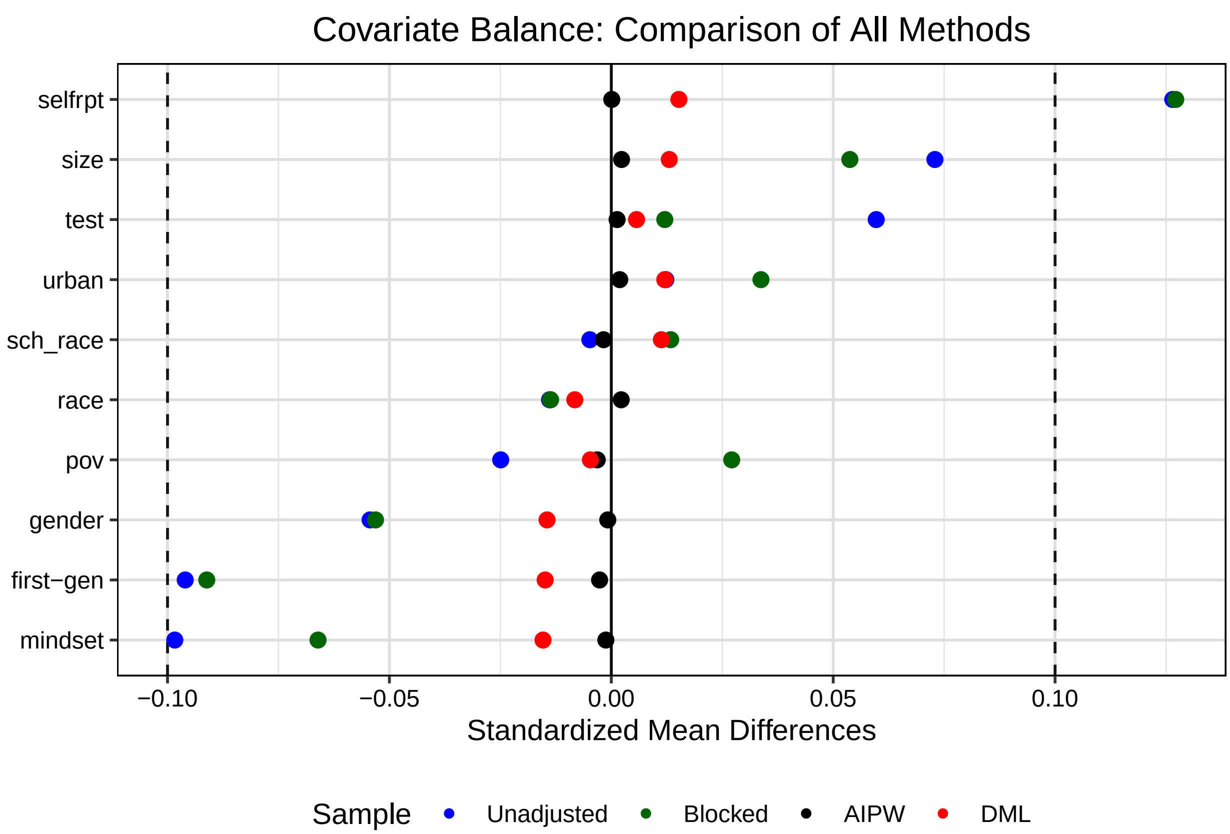
```r
# Love Plot
library(cobalt)
```

```
##  cobalt (Version 4.6.0, Build Date: 2025-04-15)
```

```r
var.names <- c(
  selfrpt   = "selfrpt",
  mindset   = "mindset",
  test      = "test",
  sch_race  = "sch_race",
  pov       = "pov",
  size      = "size",
  race_num  = "race",
  gender01  = "gender",
  urban_num = "urban",
  fgen01    = "first-gen"
)

love_plot_dots <- love.plot(
  z ~ selfrpt + mindset + test + sch_race + pov + size +
      race_num + gender01 + urban_num + fgen01,
  data         = dt,
  weights      = list(Blocked = dt$wt_block,
                      AIPW = dt$wt_aipw,
                      DML = dt$wt_dml),
  stats        = "mean.diffs",
  continuous   = "std",
  binary       = "std",
  s.d.denom    = "pooled",
  abs          = TRUE,
  var.order    = "unadjusted",
  var.names    = var.names,
  sample.names = c("Unadjusted", "Blocked", "AIPW", "DML"),
  colors       = c("black", "blue", "purple", "green"),
  shapes       = c("circle filled", "circle filled", "circle filled", "circle filled"),
  fill         = c("black", "blue", "purple", "green"),
  size         = 3,
  title        = "Covariate Balance: Comparison of All Methods",
  subtitle     = "Absolute Standardized Mean Differences",
  grid         = TRUE,
  threshold    = 0.1
) +
  theme(legend.position = "bottom")

print(love_plot_dots)
```




```r
love_plot_dots_signed <- love.plot(
  z ~ selfrpt + mindset + test + sch_race + pov + size +
      race_num + gender01 + urban_num + fgen01,
  data          = dt,
  weights       = list(Blocked = dt$wt_block,
                       AIPW = dt$wt_aipw,
                       DML = dt$wt_dml),
  stats         = "mean.diffs",
  continuous    = "std",
  binary        = "std",
  s.d.denom     = "pooled",
  abs           = FALSE,
  var.order     = "unadjusted",
  var.names     = var.names,
  sample.names = c("Unadjusted", "Blocked", "AIPW", "DML"),
  colors        = c("blue", "darkgreen", "black", "red"),
  shapes        = c(16, 16, 16, 16),
  fill          = c("blue", "darkgreen", "black", "red"),
  size          = 3,
  title         = "Covariate Balance: Comparison of All Methods",
  subtitle      = "Signed Standardized Mean Differences",
  grid          = TRUE,
  thresholds    = c(0.1, -0.1)
) +
  theme(legend.position = "bottom")

print(love_plot_dots_signed)
```




```
# Balance Metrics Calculation

bal_unadj <- bal.tab(z ~ selfrpt + mindset + test + sch_race + pov + size +
                      race_num + gender01 + urban_num + fgen01,
                    data = dt)
```

```
## Note: 's.d.denom' not specified; assuming "pooled".
```

```
unadj_metrics <- data.frame(
  Method = "Unadjusted",
  Max_SMD = max(abs(bal_unadj$Balance$Diff.Un)),
  Mean_SMD = mean(abs(bal_unadj$Balance$Diff.Un))
)

bal_block <- bal.tab(z ~ selfrpt + mindset + test + sch_race + pov + size +
                      race_num + gender01 + urban_num + fgen01,
                    data = dt, weights = dt$wt_block)
```

```
## Note: 's.d.denom' not specified; assuming "pooled".
```

```
block_metrics <- data.frame(
  Method = "Blocked",
  Max_SMD = max(abs(bal_block$Balance$Diff.Adj)),
  Mean_SMD = mean(abs(bal_block$Balance$Diff.Adj))
)
```

```r
bal_aipw <- bal.tab(z ~ selfrpt + mindset + test + sch_race + pov + size +
                      race_num + gender01 + urban_num + fgen01,
                    data = dt, weights = dt$wt_aipw)
```

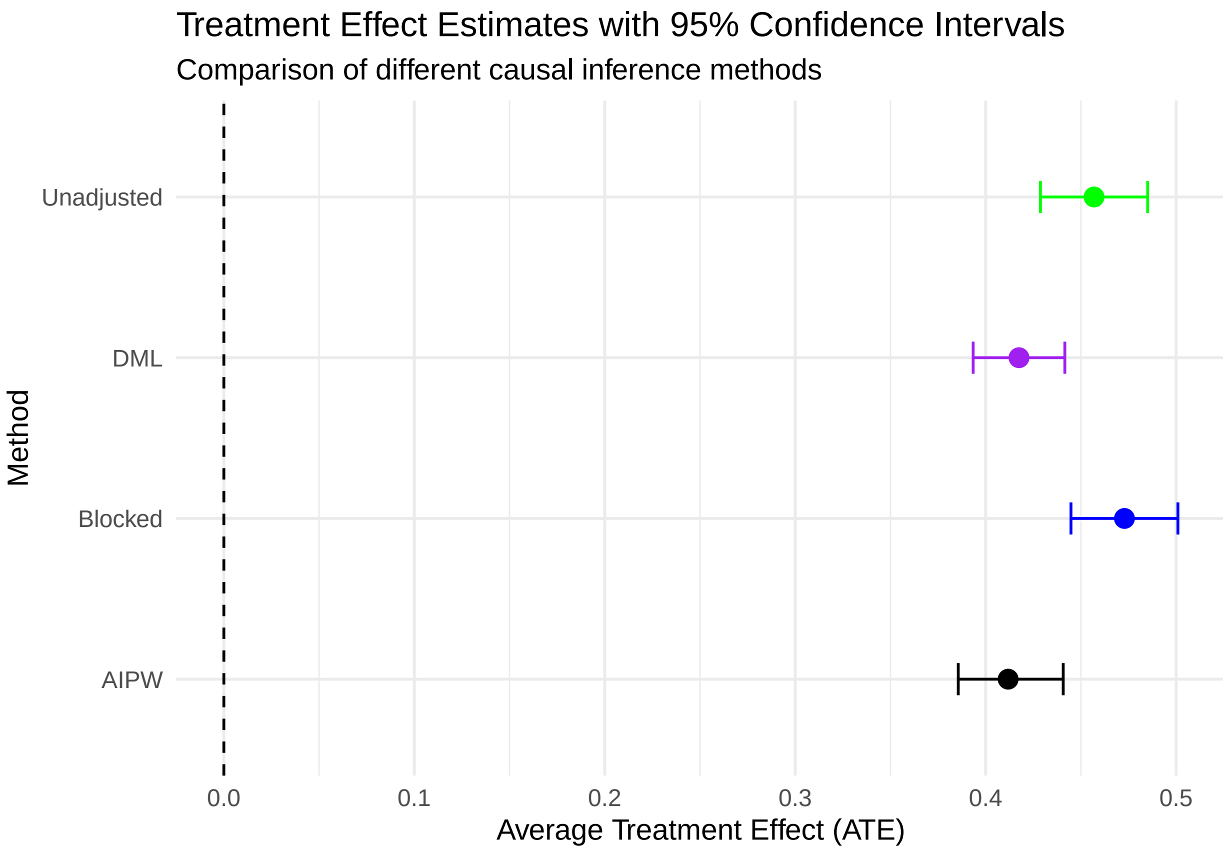
```
## Note: `s.d.denom` not specified; assuming "pooled".
```

```r
aipw_metrics <- data.frame(
  Method = "AIPW",
  Max_SMD = max(abs(bal_aipw$Balance$Diff.Adj)),
  Mean_SMD = mean(abs(bal_aipw$Balance$Diff.Adj))
)

bal_dml <- bal.tab(z ~ selfrpt + mindset + test + sch_race + pov + size +
                     race_num + gender01 + urban_num + fgen01,
                   data = dt, weights = dt$wt_dml)
```

```
## Note: `s.d.denom` not specified; assuming "pooled".
```

```r
dml_metrics <- data.frame(
  Method = "DML",
  Max_SMD = max(abs(bal_dml$Balance$Diff.Adj)),
  Mean_SMD = mean(abs(bal_dml$Balance$Diff.Adj))
)

all_metrics <- rbind(unadj_metrics, block_metrics, aipw_metrics, dml_metrics)
```

```r
treatment_effects <- data.frame(
  Method = c("Unadjusted", "Blocked", "AIPW", "DML"),
  ATE = c(unadj_est, blocked_effect, aipw_est, dml_est),
  SE = c(unadj_se, blocked_se, aipw_se, dml_se),
  CI_Lower = c(unadj_ci_lower, blocked_ci_lower, aipw_ci[1], dml_ci_lower),
  CI_Upper = c(unadj_ci_upper, blocked_ci_upper, aipw_ci[2], dml_ci_upper)
)

ggplot(treatment_effects, aes(x = ATE, y = Method, color = Method)) +
  geom_point(size = 3) +
  geom_errorbarh(aes(xmin = CI_Lower, xmax = CI_Upper), height = 0.2) +
  geom_vline(xintercept = 0, linetype = "dashed") +
  scale_color_manual(values = c("black", "blue", "purple", "green")) +
  labs(title = "Treatment Effect Estimates with 95% Confidence Intervals",
       subtitle = "Comparison of different causal inference methods",
       x = "Average Treatment Effect (ATE)") +
  theme_minimal() +
  theme(legend.position = "none")
```




```r
summary_table <- data.frame(
  Method = treatment_effects$Method,
  ATE = round(treatment_effects$ATE, 4),
  SE = round(treatment_effects$SE, 4),
  CI_Lower = round(treatment_effects$CI_Lower, 4),
  CI_Upper = round(treatment_effects$CI_Upper, 4),
  Max_SMD = round(all_metrics$Max_SMD, 4),
  Mean_SMD = round(all_metrics$Mean_SMD, 4)
)

unadj_mean_smd <- all_metrics$Mean_SMD[1]
summary_table$Bias_Reduction <- round((1 - all_metrics$Mean_SMD / unadj_mean_smd) * 100, 1)

print(summary_table)
```

```
##       Method    ATE     SE CI_Lower CI_Upper Max_SMD Mean_SMD Bias_Reduction
## 1 Unadjusted 0.4569 0.0144   0.4287   0.4851  0.1265   0.0487            0.0
## 2    Blocked 0.4729 0.0143   0.4448   0.5009  0.1272   0.0418           14.3
## 3       AIPW 0.4118 0.0137   0.3856   0.4407  0.0032   0.0016           96.8
## 4        DML 0.4175 0.0123   0.3934   0.4416  0.0154   0.0100           79.5
```